

TOPIC NAME: _____

DAY: _____

TIME: _____

DATE: / /

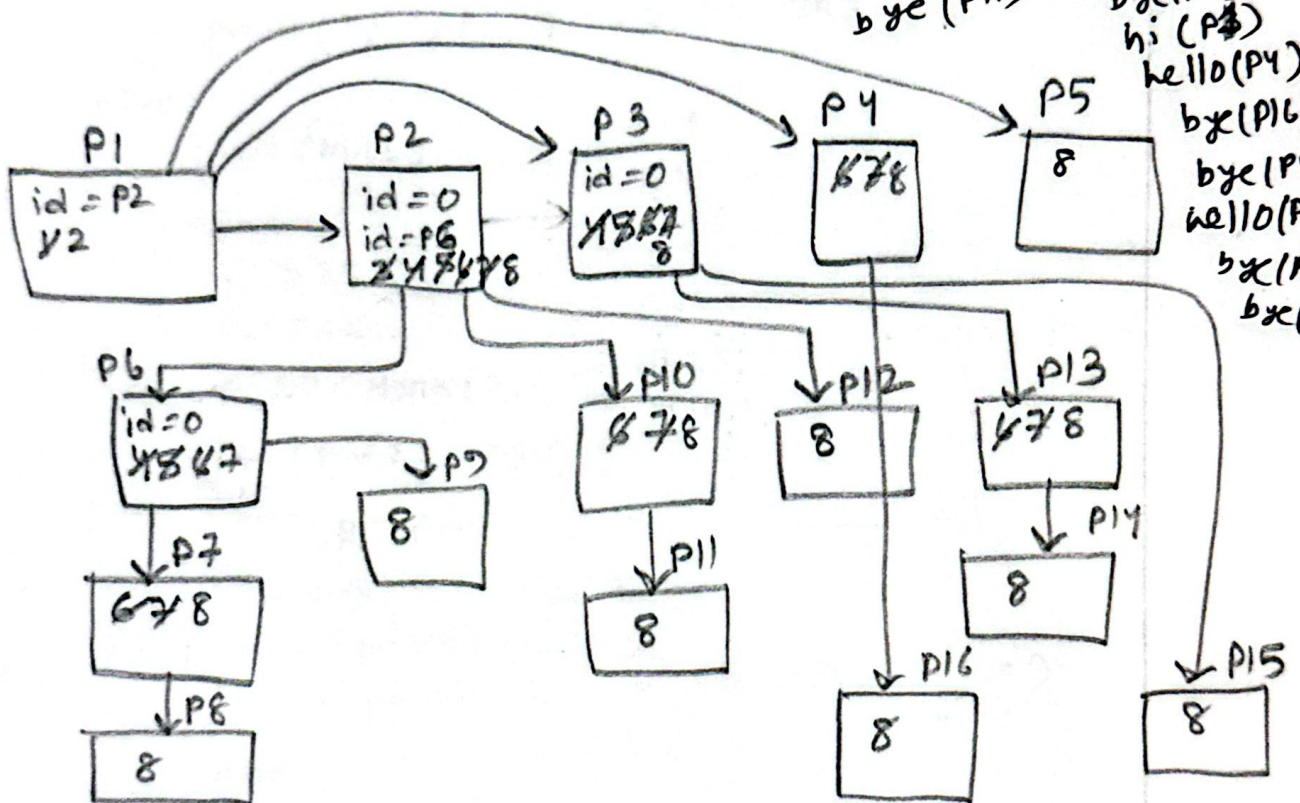
① main() { P1 (22) (26) (18)

fork(); P2
 fork(); P3
 printf("hi\n");
 fork(); P4
 printf("hello\n");
 fork(); P5
 printf("bye\n");
 }

Output

hi(P6)
 hello(P7)
 bye(P8)
 bye(P7)
 hello(P6)
 bye(P9)
 bye(P6)
 hi(P2)
 hello(P10)
 bye(P11)

bye(P16)
 hello(P2)
 bye(P12)
 bye(P2)
 hi(P3)
 hello(P13)
 bye(P14)
 bye(P13)
 hello(P3)
 bye(P5)
 bye(P3)
 hi(P4)
 hello(P4)
 bye(P16)
 bye(P4)
 hello(P4)
 bye(P5)
 bye(P1)



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(2)

```

1  main() { P1
2      fonk(); P2
3      fonk(); P3
4      c = fonk(); P4
5      if (c > 0) {
6          printf("hi\n");
7          fonk(); P5
8      }
9      fonk(); P6
10     printf("bye\n");
    }

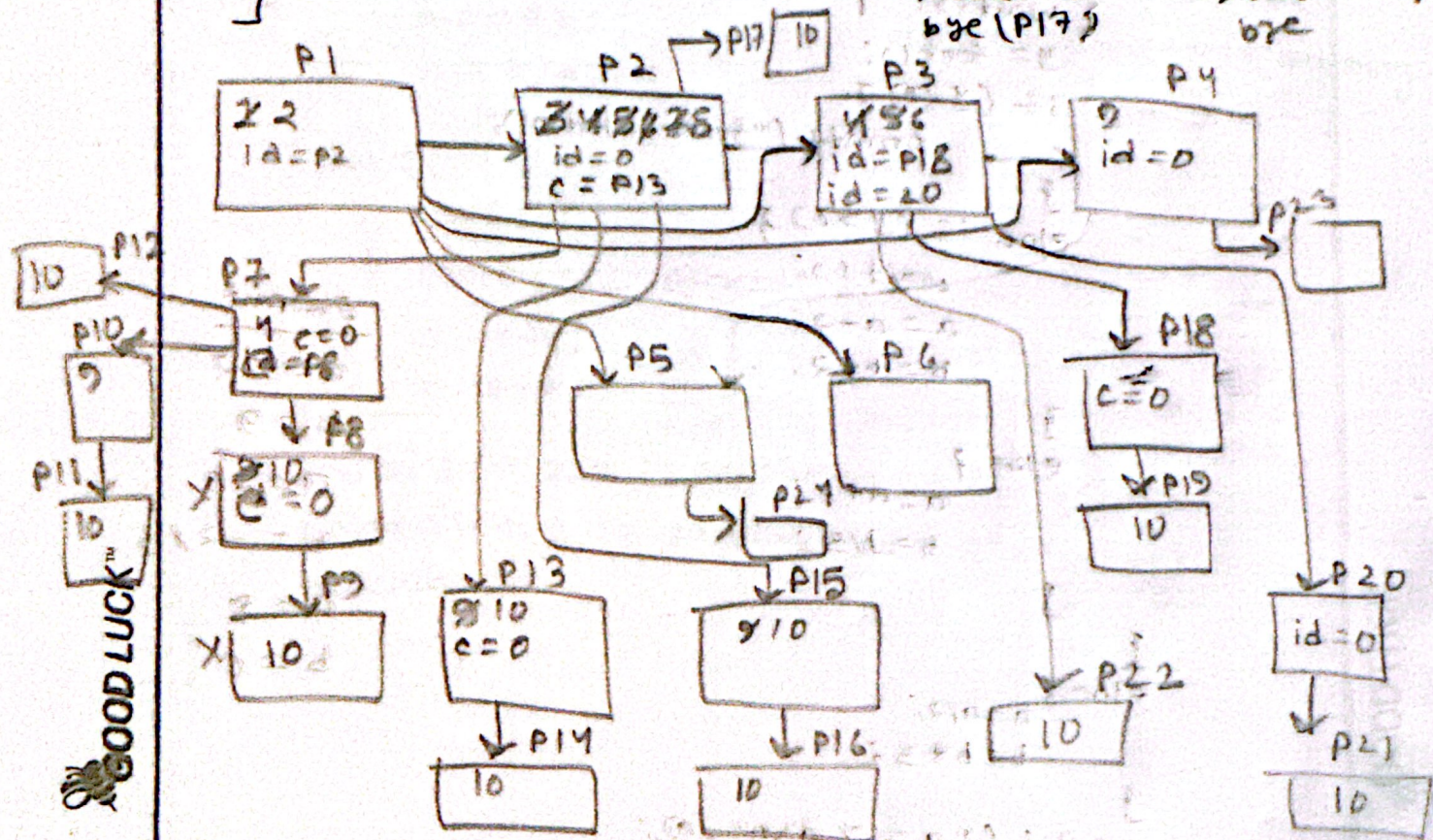
```

Output

```

bye(P9)   bye(P2)
bye(P8)   bye(P19)
hi(P7)    bye(P18)
bye(P11)  hi(P3)
bye(P10)  bye(P21)
bye(P12)  bye(P20)
bye(P7)   bye(P22)
bye(P14)  bye(P3)
bye(P13)  bye(P23)
hi(P2)    bye(P4)
bye(P16)  bye(P24)
bye(P15)  bye(P5)
bye(P17)  bye(P6)
bye

```



TOPIC NAME: _____

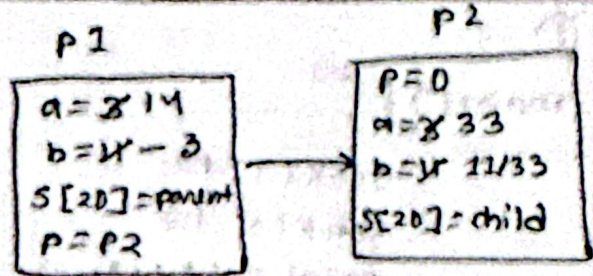
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(3)

```
int main() {
    pid_t p;
    int a = 3;
    int b = 11;
    char s[30];
    p = fork();
    if (p < 0) {
        printf("fork failed\n");
    }
    else if (p == 0) {
        strcpy(s, "child");
        a = a + b;
        b = b / a;
    }
    else {
        wait();
        strcpy(s, "parent");
        a = a + b;
        b = b - a;
    }
    printf("%s is printing a = %d\n", s, a);
    printf("%s is printing b = %d\n", s, b);
    return 0;
}
```



output:

child is printing

a = 33

child is printing

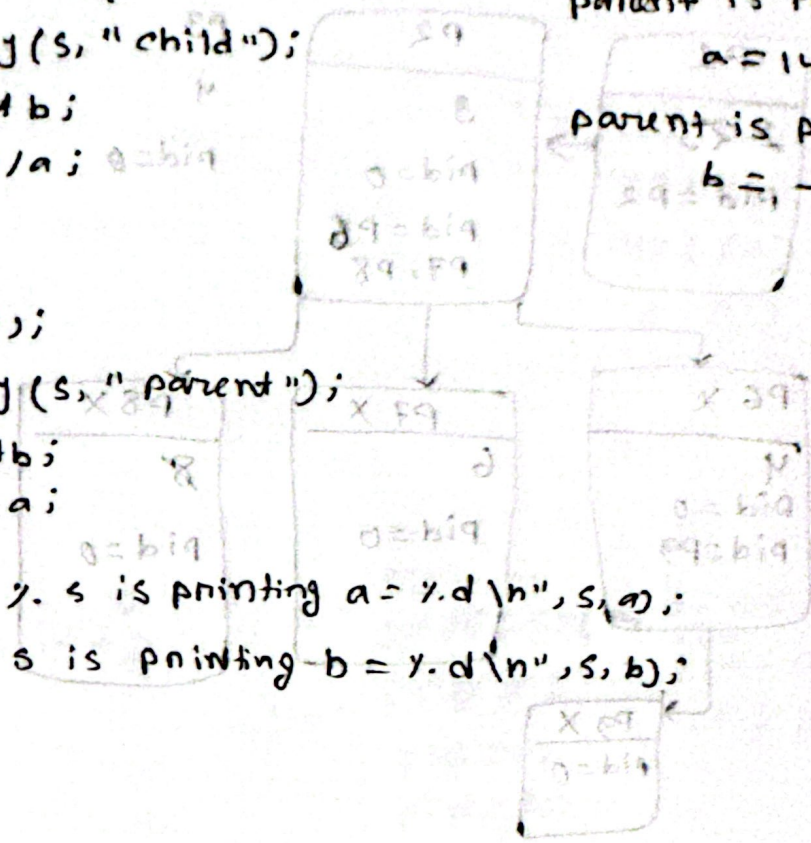
b = 11/33

parent is printing

a = 14

parent is printing

b = -3



GOOD LUCK™

TOPIC NAME: _____

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(4)

```
static int a = 5;
static int b = 3;
int main() {
    pid_t x, y;
    x = fork();
    if (x < 0) {
        printf("fork failed\n");
    }
    else if (x > 0) {
        a = a + 5;
        b = b - 5;
        wait();
        y = fork();
        if (y < 0) {
            printf("fork failed\n");
        }
        else if (y > 0) {
            wait();
            a = a - 2;
            b = b + 2;
        }
        else {
            a = a * 2;
            b = b / 3;
        }
    }
    else {
        a = a / 2;
        b = b * 3;
    }
    printf("a = %d\n", a);
    printf("b = %d\n", b);
    return 0;
}
```

P1

a = 8

b = 3 - 20

x = p2

y = p3

P2

a = 5/2

b = 3/9

x = 0

P3

a = 10/20

b = -2/3

x = p2

y = 0

Output

a = 5/2

b = 9

a = 20

b = -2/3

a = 8

b = 0

GOOD LUCK™

Address

TOPIC NAME: Quiz 1 (Set A)

DAY:

TIME:

DATE: / /

①

```
int x = 10;
if (fork() > 0)
{
    x = x + 20;
    b = fork();
    wait_pid(b);
}
else
{
    x = x - 20;
    printf(x);
}
wait();
printf(x);
```

P1
x = 30
id = P2
b = P3

P2
x = 10 - 10
id = 0

P3
x = 30
b = 0

Output

-10 (P2)
-10 (P2)
30 (P3)
30 (P1)

GOOD LUCK

TOPIC NAME: Quiz 1 (B set) (10/2) DAY: 2

TIME:

DATE:

```
int y = 20;
if (fork() > 0)
{
    y = y - 20;
    b = fork();
    waitpid(b, 0);
}
else
{
    y = y + 20;
    printf(y);
}
```

wait();
printf(y);

Output

(20) 01-

(20) 01-

(20) 01-

```
1 int y = 20;
2 for (i = 0; i < 2; i++)
3 {
4     y = y + 3;
5     fork();
6 }
7 wait();
8 printf(y);
```

P1
y = 200
pid = P2
b = P3

P2
y = 2040
pid = 0

P3
y = 0
b = 0

Output

40 (P2)

40 (P2)

0 (P3)

0 (P1)

P1
y = 202526
i = 2
pid = P2
pid = P3

P2
y = 2826
pid = 0
i = 2

P3
y = 26
i = 2

P4
y = 26
i = 2